

ATHARVA HEMANT JOSHI

+91 8999-621-553 | josh.atharva2005@gmail.com | www.linkedin.com/in/atharva-joshi-35a254296/ | Pune, Maharashtra, India

TECHNICAL SKILLS

Languages: Python, Java, C++, SQL, Shell Scripting , JavaScript

Java / J2EE Stack: Servlets, JDBC, Struts, EJB, Android SDK (Java), RESTful Web Services

Web Technologies: React.js, FastAPI, Flask, WebSockets , REST API integration, PHP

AI & Agentic Software Development: Agentic Task Pipelines, Model Orchestration, Inference Pipelines , OpenCV, YOLO, MLFlow, Scikit-learn, TensorFlow , ETL Pipelines

Data & Cloud: MongoDB, MySQL, AWS (S3, EC2, DynamoDB), Docker , Git , Redis , CI/CD, Linux

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, System Design (Fundamentals)

EXPERIENCE

Computer Vision & ML Intern | *Thelios AI*

Jun 2025 - Sep 2025

- Engineered Python-based ETL pipelines for large-scale image datasets cutting preprocessing latency by 35% via batched ingestion and parallelized transforms; integrated REST API endpoints for pipeline control.
- Fine-tuned YOLO deep learning models with custom augmentation pipelines, improving mAP by 18% over baseline; tracked experiments with MLflow, wrote unit tests, and shipped reproducible builds end-to-end.

Research Lead Intern - AI/ML | *AWS Cloud Club - PICT*

Feb 2025 - Jun 2025

- Delivered a working ML classification system at 82% accuracy, measured against held-out test data, by leading a 6-member team through the full design-build-evaluate cycle. Standardized the team's engineering process into a documented, reusable framework, measured by adoption across all 6 team members, by authoring a faculty-reviewed methodology report.

PROJECTS

ISRO IRoC-U Robotics Challenge 2025 - Autonomous Drone Mission Control

Python, Flask (REST APIs), HTTP Server, ROS, OpenCV, LiDAR, NumPy | *AIR 1 / 500+ Teams - IITs, NITs*

- Architected a full-stack autonomous drone system with a Flask REST API backend and real-time web dashboard; designed a novel 2D→2.5D LiDAR terrain-mapping algorithm using statistical point-cloud filtering, achieving sub-metre GPS-free navigation accuracy.
- Built modular OpenCV inference pipelines for real-time object detection and flatness scoring; shipped clean, testable Python services with clearly structured API contracts across each subsystem.

ISRO BAH 2025 - Satellite Air Pollution Analytics

Python, Pandas, Scikit-learn, NumPy, Matplotlib | *Top 28 / 8,744 Teams, 160K+ Registrations*

- Designed an end-to-end data pipeline ingesting 5+ years of multi-source satellite AQI data; applied feature engineering to distil 12 high-signal variables from 40+ raw inputs and shipped regression/classification models hitting 88% accuracy with F1 improved 15% via hyperparameter tuning.

Green Model Advisor - Carbon-Aware AI Inference Router

Python, FastAPI, Docker, Proprietary API, CodeCarbon

- Developed an AI inference routing platform that dynamically selects LLMs based on latency, accuracy, cost, and estimated CO₂ emissions using FastAPI microservices.
- Integrated real-time carbon tracking with Dockerized deployment, reducing compute usage by **45%** compared with static model selection.

CollabIQ - AI-Powered Project Management Platform

ReactJS • FastAPI • MySQL • Proprietary API's • Docker

- Built a full-stack SaaS platform featuring AI-assisted task management, documentation generation, collaborative coding, analytics dashboards, and role-based authentication. Designed scalable REST APIs, modular backend services, and LLM-powered automation to streamline project planning and improve developer productivity.

SHETISAKHA — Agri-Bot Control System (Java / Android)

Java, Android 14, HC-05 Bluetooth, Sensor Integration

- Built an Android 14-compatible control app that transmits real-time soil moisture, pH, and nutrient data, measured by reliable sensor-to-app data sync, by integrating an onboard soil-monitoring sensor package over Bluetooth.
- Enabled autonomous field-path replay, measured by accurate trajectory reproduction on live hardware, by implementing a Java path-memorizing algorithm (record/play via HC-05 Bluetooth) plus a touch-based Draw Lane navigation mode.
- Delivered core Agri-Bot task automation, measured by successful field trials, by coordinating app-to-hardware control for automated seeding, sprinkling, and ploughing.

EDUCATION

B.E. Computer Engineering - Pune Institute of Computer Technology (PICT) (*Lateral Entry, 2nd year, via Diploma*) Aug 2024 - Jul 2027

Diploma in Information Technology - Government Polytechnic, Nagpur

Sep 2021 - May 2024

ACHIEVEMENTS & LEADERSHIP

- AIR 1 - ISRO IRoC-U Robotics 2025: Ranked 1st nationwide (500+ teams incl. IITs & NITs) for GNSS-free autonomous drone with LiDAR-based CV.
- AIR 9 - DD Robocon 2026: 100/100 in Stage 1 & Stage 2; Best Technical Report Award, highest technical evaluation score.
- Top 28 - ISRO Bharat Antariksh Hackathon 2025 (8,744 teams): ML-based AQI prediction pipeline.
- Finance & Project Management Lead 2026, PICT Robotics Club: Leading a 30-member team; owns architecture, task pipelines, and integration milestones.